THE FITTING IDEALS OF A FINITE TYPE MODULE REPRESENT THE LOCI WHERE IT IS GENERATED BY R SECTIONS AND WHERE IT IS LOCALLY FREE OF RANK R

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Following the multi-claim convention for this source, source_url is the source-level URL https://stacks.math.columbia.edu/ and the source label lemma-locally-free-rank-r-pullback is carried in claim_key and here rather than in a per-claim tag URL.

Editorial note. The preamble is the chapter's shared Stacks Project preamble, retained as sources/stacks-divisors/retained-preamble.tex: a line-for-line excerpt of the shared Stacks preamble staged with this snapshot. No macro was invented or redefined, and this card adds no preamble text of its own.

Editorial note. Statement span 1383-1422 and proof span 1424-1456 come from the halfA pool card operations/pool-cards/stacks-divisors-halfA.json and were re-read in the pinned chapter file before this card was written.

Editorial note. The statement contains a large displayed piecewise functor definition with a matrix-style layout; it is copied verbatim from the counted statement span.

Editorial note. 7 cross-references occur inside the counted spans; they are listed in external_references as navigation pointers only.

Allowed standard prerequisites (editorial).

- Fitting ideals of a finite type module and their behaviour over affine opens, as cited from the more-algebra chapter, including that over an affine open the n-th Fitting ideal becomes the whole ring for large n. Cited at proof lines 1425-1427. Gives part (1), the emptiness of the intersection of the closed subschemes defined by the Fitting ideals.
- Base change of Fitting ideals and the same-chapter criteria for a module to be locally generated by r sections, respectively to be finite locally free of rank r, in terms of the r-th and the (r-1)-st Fitting ideal. Cited at proof lines 1430-1434 and 1439-1444. Converts the functorial statements (2) and (3) into statements about the vanishing of the pullback of a Fitting ideal.
- For a finitely presented module the Fitting ideals are of finite type, so the closed immersions they cut out are of finite presentation, and a retrocompact open in the image of such an immersion is of finite presentation. Cited at proof lines 1448-1455. Gives the final finite presentation assertions about the three morphisms.

Source reference. Original lemma 0C3D: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 15188–15188. Same-chapter reference used in the retained author text at source line(s) 1432, 1440. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's

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Source reference. Original lemma 0C3G: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 15191–15191. Same-chapter reference used in the retained author text at source line(s) 1441, 9362, 9367, 9372. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 0C3F: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 15190–15190. Same-chapter reference used in the retained author text at source line(s) 1433, 9317. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 0C3E: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 15189–15189. Same-chapter reference used in the retained author text at source line(s) 1450, 9360. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original section more-algebra-section-fitting-ideals: sources/stacks-divisors/pinned-original-source.tex, source lines 1251–1251. Same-chapter reference used in the retained author text at source line(s) 1251, 1428. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 084Q: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10330–10330. Same-chapter reference used in the retained author text at source line(s) 1451, 5010, 5119, 8124, 8679, 9099. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma properties-lemma-quasi-coherent-finite-type-ideals: sources/stacks-divisors/pinned-original-source.tex, source lines 1453–1453. Same-chapter reference used in the retained author text at source line(s) 1453, 8683, 9187. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Lemma 0.1. *Let S be a scheme. Let $\mathcal{F}$ be a finite type, quasi-coherent $\mathcal{O}_S$-module. The closed subschemes*

$$S = Z_{-1} \supset Z_0 \supset Z_1 \supset Z_2 \ldots$$

defined by the Fitting ideals of $\mathcal{F}$ have the following properties

(1) *The intersection $\bigcap Z_r$ is empty.*
(2) *The functor $(Sch/S)^{opp} \to Sets$ defined by the rule*

$$T \longmapsto \begin{cases} \{*\} & \text{if } \mathcal{F}_T \text{ is locally generated by } \leq r \text{ sections} \\ \emptyset & \text{otherwise} \end{cases}$$

is representable by the open subscheme $S \setminus Z_r$.

(3) *The functor* $F_r : (Sch/S)^{opp} \to Sets$ *defined by the rule*

$$T \longmapsto \begin{cases} \{*\} & \text{if } \mathcal{F}_T \text{ locally free rank } r \\ \emptyset & \text{otherwise} \end{cases}$$

is representable by the locally closed subscheme $Z_{r-1} \setminus Z_r$ *of* S.

If $\mathcal{F}$ *is of finite presentation, then* $Z_r \to S$, $S \setminus Z_r \to S$, *and* $Z_{r-1} \setminus Z_r \to S$ *are of finite presentation.*

Proof. Part (1) is true because over every affine open U there is an integer n such that $\mathrm{Fit}_n(\mathcal{F})|_U = \mathcal{O}_U$. Namely, we can take n to be the number of generators of $\mathcal{F}$ over U, see More on Algebra, Section more-algebra-section-fitting-ideals.

For any morphism $g : T \to S$ we see from Lemmas 0C3D and 0C3F that $\mathcal{F}_T$ is locally generated by $\leq r$ sections if and only if $\mathrm{Fit}_r(\mathcal{F}) \cdot \mathcal{O}_T = \mathcal{O}_T$. This proves (2).

For any morphism $g : T \to S$ we see from Lemmas 0C3D and 0C3G that $\mathcal{F}_T$ is free of rank r if and only if $\mathrm{Fit}_r(\mathcal{F}) \cdot \mathcal{O}_T = \mathcal{O}_T$ and $\mathrm{Fit}_{r-1}(\mathcal{F}) \cdot \mathcal{O}_T = 0$. This proves (3).

Assume $\mathcal{F}$ is of finite presentation. Then each of the morphisms $Z_r \to S$ is of finite presentation as $\mathrm{Fit}_r(\mathcal{F})$ is of finite type (Lemma 0C3E and Morphisms, Lemma 084Q). This implies that $Z_{r-1} \setminus Z_r$ is a retrocompact open in Z_r (Properties, Lemma properties-lemma-quasi-coherent-finite-type-ideals) and hence the morphism $Z_{r-1} \setminus Z_r \to Z_r$ is of finite presentation as well. □